Sokoine University of Agriculture (SUA)
Department of Agricultural Engineering
Electronics and Precision Agriculture Lab (EPAL)

AGE 418
Sensors and Controls in Precision Agriculture

Lecture 13 Compendium:
Agricultural Robotics and Automation
The Future of Farming: From Automation to Autonomy

Instructor: Dr. Kadeghe Fue
Academic Year: 2025

Contents

1 1. Introduction: Why Robots? 2
1.1 The 3 D's of Robotics . . . 2

2 2. Levels of Autonomy 2

3 3. Robot Architecture: Sense-Think-Act 2
3.1 1. Perception (Sense) . . . 3
3.2 2. Processing (Think) . . . 3
3.3 3. Actuation (Act) . . . 3

4 4. Weeding Robots 3
4.1 Mechanical Weeding . . . 3
4.2 Chemical Weeding (Spot Spraying) . . . 3
4.3 Thermal/Laser Weeding . . . 4

5 5. Harvesting Robots 4
5.1 Challenges . . . 4
5.2 Solutions . . . 4

6 6. Unmanned Aerial Vehicles (UAVs / Drones) 4
6.1 Scouting Drones (Fixed Wing) . . . 4
6.2 Spraying Drones (Multi-Rotor) . . . 4

7 7. Swarm Robotics 5
7.1 The Concept . . . 5
7.2 Challenges of Swarms . . . 5

8 8. Automated Tractor-Implement Management (TIM) 5

9 9. Lecture Assessment 6
9.1 Part A: Standard Concept Questions . . . 6
9.2 Part B: Advanced Analysis (Open Source Data) . . . 6

1 1. Introduction: Why Robots?

Agriculture is facing a global labor crisis. Young people are moving to cities, and the remaining workforce is aging. Simultaneously, we need to produce more food with fewer chemicals and less compaction.

1.1 The 3 D's of Robotics

Robots are ideally suited for tasks that are: 1. **Dull:** Repetitive tasks (e.g., weeding row after row for 10 hours). 2. **Dirty:** Tasks involving soil, dust, or chemicals (e.g., spraying pesticides). 3. **Dangerous:** Tasks with high risk of injury (e.g., climbing tall fruit trees or working in extreme heat).

Automation vs. Autonomy

- **Automation:** A machine does a specific task automatically, but a human supervises closely (e.g., a combine harvester maintaining constant RPM).
- **Autonomy:** The machine perceives its environment, makes decisions, and acts without human intervention (e.g., a robot navigating a field and deciding which plant is a weed).

2 2. Levels of Autonomy

Just like self-driving cars, agricultural robots are classified by levels of independence.

- **Level 0 (No Automation):** Driver steers, shifts gears, and controls implement. (Old Massey Ferguson 135).
- **Level 1 (Driver Assistance):** Cruise control or simple lightbar guidance.
- **Level 2 (Partial Automation):** Auto-Steer. The tractor steers itself, but the driver must sit in the seat and turn at headlands.
- **Level 3 (Conditional Automation):** Tractor handles headland turns and implement control (ISOBUS Class 3). Driver is present but "hands-off."
- **Level 4 (High Automation):** Supervised Autonomy. One driver supervises a "Leader" tractor while a "Follower" tractor (driverless) mimics it (Grain Cart sync).
- **Level 5 (Full Autonomy):** No cab, no driver. The robot is deployed and works entirely alone (e.g., Swarm Robots).

3 3. Robot Architecture: Sense-Think-Act

Every agricultural robot consists of three main subsystems loop.

3.1 1. Perception (Sense)

The robot must understand its environment.

- **GNSS (RTK):** For absolute positioning ("I am at coordinate X, Y").
- **LiDAR:** Lasers scan for obstacles (people, trees) and navigation.
- **Cameras:** Computer Vision to identify crop rows or weeds.
- **IMU (Inertial Measurement Unit):** Detects tilt/roll on uneven terrain.

3.2 2. Processing (Think)

The "Brain" (Edge Computing).

- **Path Planning:** Calculating the most efficient route to cover the field.
- **Object Detection:** "Is that green object a maize plant or a weed?"
- **Decision Logic:** "If weed, activate sprayer."

3.3 3. Actuation (Act)

The "Muscles."

- **Locomotion:** Electric motors or hydraulic wheels for movement.
- **Manipulation:** Robotic arms (fruit picking) or Solenoid valves (spraying).

4 4. Weeding Robots

Weeding is the "Killer App" for ag robots because herbicides are expensive and resistance is growing.

4.1 Mechanical Weeding

- **Technology:** Cameras detect the crop row. Hydraulic blades slide in and out *between* plants to cut weeds without damaging the crop.
- *Example:* Garford Robocrop.

4.2 Chemical Weeding (Spot Spraying)

- **Technology:** "See & Spray." Cameras identify weeds using AI. Micro-nozzles spray herbicide *only* on the weed.
- *Impact:* Reduces chemical usage by 90% compared to blanket spraying.
- *Example:* Blue River Technology (John Deere).

4.3 Thermal/Laser Weeding

- **Technology:** High-power lasers zap the meristem (growth point) of the weed, boiling the cells instantly.
- *Pros:* No chemicals, no soil disturbance (good for regenerative ag).
- *Example:* Carbon Robotics.

5 5. Harvesting Robots

Harvesting is the hardest challenge because biological products are soft, variable, and easily bruised.

5.1 Challenges

1. **Occlusion:** Fruit is often hidden behind leaves.
2. **Variability:** No two apples are the same size or shape.
3. **Delicacy:** Using metal grippers bruises fruit.

5.2 Solutions

- **Soft Robotics:** Using silicone/rubber grippers filled with air to gently grab fruit.
- **Vacuum Effect:** Suction cups to hold fruit while a blade cuts the stem.
- **Stereo Vision:** Using two cameras to perceive depth (3D) to locate the fruit in space.

6 6. Unmanned Aerial Vehicles (UAVs / Drones)

Drones are "Flying Robots."

6.1 Scouting Drones (Fixed Wing)

- *Use:* Mapping large areas (500+ ha).
- *Sensors:* Multispectral (NDVI) or RGB.
- *Flight Time:* Long (45-90 mins).

6.2 Spraying Drones (Multi-Rotor)

- *Use:* Application of pesticides/fertilizers.
- *Capacity:* 10L - 40L tanks.
- *Physics:* The downdraft from the propellers pushes the chemical deep into the canopy, improving coverage.
- *Advantage:* Can spray over wet paddy fields (e.g., Mbarali rice farms) or steep hills (Uluguru mountains) where tractors cannot go.

7 7. Swarm Robotics

Instead of one giant 500HP tractor (which causes soil compaction), why not use 10 small 50HP robots?

7.1 The Concept

A "Swarm" of small robots works together to finish a field.

- **Redundancy:** If one robot breaks down, the other 9 keep working. The job gets done. If a big tractor breaks, the whole farm stops.
- **Compaction:** Small robots are light. They do not crush soil pores.
- **Scalability:** A farmer can buy 1 robot this year, and another next year. Easier entry cost.

7.2 Challenges of Swarms

- **Logistics:** How do you refuel/recharge 10 robots?
- **Theft:** Small robots are easier to steal in unsecured fields.

8 8. Automated Tractor-Implement Management (TIM)

This is **ISOBUS Class 3**. Normally, the Tractor controls the Implement. With TIM, the **Implement controls the Tractor**.

- *Example:* A Baler. As the bale chamber fills up, the *Baler* tells the *Tractor* to "Stop." The Baler wraps and ejects the bale. Then the Baler tells the Tractor to "Go."
- The driver effectively does nothing but supervise safety.

9 9. Lecture Assessment

9.1 Part A: Standard Concept Questions

1. List the "3 D's" that make a task suitable for agricultural robotics.
2. Differentiate between **Automation** and **Autonomy**. Give an agricultural example of each.
3. Explain the three main subsystems of a robot (The Sense-Think-Act loop).
4. What is **ISOBUS Class 3 (TIM)**? How does it differ from standard ISOBUS?
5. Compare **Green-on-Brown** vs. **Green-on-Green** weed detection. Which one requires Artificial Intelligence (AI)?
6. List three advantages of using **Spray Drones** over traditional backpack or tractor spraying in Tanzania.
7. What is **Swarm Robotics**? How does it solve the problem of soil compaction?
8. Explain the challenge of **Occlusion** in robotic fruit harvesting.
9. What is **Soft Robotics**, and why is it necessary for harvesting tomatoes or strawberries?
10. Identify two major barriers to the adoption of autonomous tractors in rural Africa.

9.2 Part B: Advanced Analysis (Open Source Data)

These questions emphasize the use of open-source datasets/tools and engineering logic.

1. **Robot Path Planning Logic:** A robot needs to cover a rectangular field of Maize. *Task:* Draw/Describe a "Boustrophedon" (Ox-turning) path. Write a pseudocode logic for the turning event at the headland. *Inputs:* GPS Heading, Field Boundary Geofence. *Logic:* IF `Distance_To_Boundary` < 2m THEN `Slow_Down` AND `Turn_180_Degrees`.
2. **Drone Spraying Calculation:** You are a service provider using a DJI Agras T30 (30L tank). *Task:* You need to spray a 10-hectare rice paddy in Kilombero. The application rate is 15 Liters/ha.
   - How many total liters are needed?
   - How many refill flights will the drone make?
   - If one battery lasts 10 minutes and covers 1 hectare, how many batteries do you need to charge to finish the job non-stop?
3. **Visualizing Swarm Efficiency:** Use **Python** or Excel to model work rates. *Task:*
   - **Scenario A:** One large tractor (width 12m, speed 10km/h).
   - **Scenario B:** Four small robots (width 3m each, speed 5km/h).

   Calculate the theoretical field capacity (ha/hr) for both. If one small robot fails, what is the new capacity of the swarm? If the big tractor fails, what is the capacity?

4. **Safety Lidar Analysis:** Search for **Velodyne** or **Sick** LiDAR sensor specs online. *Task:* Look at the "Field of View" and "Range" specs. If an autonomous tractor travels at 10 km/h, and it takes 5 meters to stop instantly upon braking, what is the *minimum* detection range required for the LiDAR to safely detect a human/cow in front of it?

5. **Open Source Robotics (ROS):** Search for **ROS Agriculture** (Robot Operating System). *Task:* Identify one open-source package available for agricultural navigation (e.g., `ag_stack` or `gps_waypoint_nav`). Explain how using Open Source software reduces the development cost for a Tanzanian startup building a weeding robot.